
Online Examination Report

Date:	20.08.2014
Attempts:	1
Examinee:	Okirol, Samuel
Date of birth:	
Subject:	020-AGK
Result:	38,99 %
Time used:	00:00:00 (HH:MM:SS)

Attachments

1. List of result by questions
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by questions

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	694	22.2.4	Altimeter	0,00	1,00
2	759	21.9.3.1	DC Generation	0,00	1,00
3	2262	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
4	2399	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
5	1422	22.6.3	Flight Director : design and operation.	1,00	1,00
6	4122	22.	INSTRUMENTATION	1,00	1,00
7	1054	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
8	1954	22.2.4	Altimeter	0,00	1,00
9	2044	22.9	AUTOTHROTTLE	1,00	1,00
10	2422	21.2.3.3	Loads, stresses flutter	0,00	1,00
11	2297	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
12	2448	21.1	SYSTEM DESIGN, LOADS, STRESSES, MAINTENANCE	0,00	1,00
13	546	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
14	2288	21.9.1	General, definitions, basic applications: etc	0,00	1,00
15	355	22.2.1	Pressure measurement	1,00	1,00
16	1237	22.2.6	Airspeed indicator (ASI)	1,00	1,00
17	1938	22.2.4	Altimeter	1,00	1,00
18	2781	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
19	662	22.2.1	Pressure measurement	1,00	1,00
20	142	22.2.1	Pressure measurement	1,00	1,00
21	1829	22.2.6	Airspeed indicator (ASI)	0,00	1,00
22	1919	22.2.4	Altimeter	0,00	1,00
23	3871	22.	INSTRUMENTATION	0,00	1,00
24	1154	21.9.1.1	Static electricity	0,00	1,00
25	1812	22.6.3	Flight Director : design and operation.	0,00	2,00
26	2461	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
27	2020	21.9.1	General, definitions, basic applications: etc	1,00	1,00
28	1837	22.5.2	IRS: (Strapped-down)	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	2769	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
30	1374	22.2.7	Mach meter	0,00	1,00
31	343	21.13	OXYGEN SYSTEMS	0,00	1,00
32	574	22.6.3	Flight Director : design and operation.	1,00	1,00
33	619	21.1.1.1	Design concepts	0,00	1,00
34	1091	21.13	OXYGEN SYSTEMS	0,00	1,00
35	1915	22.12.8	Radio Altimeter	0,00	1,00
36	1105	21.8.2.2	Design, operation, system components, etc	0,00	1,00
37	2636	22.	INSTRUMENTATION	1,00	1,00
38	741	21.2.3	Aeroplane: Wings, tail and control surfaces	1,00	1,00
39	197	22.2.1	Pressure measurement	0,00	1,00
40	2367	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	0,00	1,00
Total:39%				16,00	41,00

Attachment 2: List of result by topic

Licence

Date: 20.08.2014

Examinee: Okirol, Samuel

Location: Uganda Civil Aviation Authority

Your Result: 16,00 from 41,00 Points (39,02 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	6,00	20,00	30,00
21.1. SYSTEM DESIGN, LOADS, STRESSES, MAINTENANCE	2	0,00	2,00	0,00
21.2. AIRFRAME	5	1,00	5,00	20,00
21.4. LANDING GEAR, WHEELS, TYRES, BRAKES	1	0,00	1,00	0,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	3	3,00	3,00	100,00
21.8. FUEL SYSTEM	1	0,00	1,00	0,00
21.9. ELECTRICS	4	1,00	4,00	25,00
21.13. OXYGEN SYSTEMS	2	0,00	2,00	0,00
22. INSTRUMENTATION	20	10,00	21,00	47,62
22.2. MEASUREMENT OF AIR DATA PARAMETERS	11	5,00	11,00	45,45
22.5. INS and IRS	1	0,00	1,00	0,00
22.6. AEROPLANE : AUTOMATIC FLIGHT CONTROL SYSTEMS	3	2,00	4,00	50,00
22.9. AUTOTHROTTLE	1	1,00	1,00	100,00
22.12. ALERTING SYSTEMS, PROXIMITY SYSTEMS	1	0,00	1,00	0,00
Total	40	16,00	41,00	39,02

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1

The high Altitude Radio Altimeter uses the following wavelengths:

☐ **decimetric.**

☐ **myriametric.**

☐ **hectometric.**

☒ **metric.**

2

The voltage regulator of a DC generator is connected in:

- ☐ *series with the equalising coil.*
- ☐ *series with the shunt field coil.*
- ☒ *parallel with the shunt field coil.*
- ☐ *series with the armature.*

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3

The cabin air for modern airplanes is usually supplied by:

- ☐ piston compressors.
- ☐ single radial compressors.
- ☐ roots type compressors.
- ☒ main engine compressors.

4

What is the purpose of an anti-servo tab?

- ☒ **Reduce control forces by deflecting in the proper direction to move a primary flight control.**
- ☐ **Prevent a control surface from moving to a full-deflection position due to aerodynamic forces.**
- ☐ **Move the flight controls in the event of manual reversion.**

5

The horizontal command bar of a flight director:

- 1 - repeats the position information given by the ILS in the horizontal plane
- 2 - repeats the position information given by the ILS in the vertical plane
- 3 - gives information about the direction and the amplitude of the corrections to be applied on the pitch of the aircraft.

The combination that regroups all of the correct statements is:

☐ 1, 3.

☒ 3.

☐ 2, 3.

☐ 2.

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Date of birth: --

Subject: 020-AGK
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6

(Refer to Figure 84.) Which altimeter depicts 8,000 feet?

See Attachments:

1. instrument_84.jpg

Figure 84

(Attachment
No.1)

☐ 1

☒ 3

☐ 2

7

The wing of an aircraft in flight, powered by engines located under the wing, is subjected to a bending moment which causes its leading edge, from the wing root to the wing tip, to operate in:

☐ compression, then in tension.

☒ tension.

☐ compression.

☐ tension, then in compression.

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8

When ambient temperature is warmer than standard at a particular altitude, the altimeter will indicate:

- ☐ lower than true altitude;
- ☒ higher than true altitude;
- ☐ the same as true altitude.

9

During a climb with the autopilot engaged in the V/S mode (holding of vertical speed), the auto-throttle:

- ☐ can be operated in the *SPEED* (holding of IAS) or *N1* (holding of N1) modes.
- ☐ can be operated in the *N1 / EPR* mode (holding of N1 or EPR).
- ☒ can be operated in the *SPEED* mode (holding of IAS).
- ☐ is not available.

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10

Upon which factor does wing loading during a level coordinated turn in smooth air depend?

☐ Angle of bank.

☐ True airspeed.

☒ Rate of turn.

11

Assuming cabin differential pressure has attained the required value in normal flight conditions, if flight altitude is maintained:

- ☒ a constant mass air flow is permitted through the cabin.
- ☐ the outflow valves will move to the fully closed position.
- ☐ the pressurisation system ceases to function until leakage reduces the pressure.
- ☐ the outflow valves will move to the fully open position.

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12

Which maximum range factor decreases as weight decreases ?

☒ Angle of attack.

☐ Airspeed.

☐ Altitude.

13

The term "Bootstrap", when used to identify a cabin air conditioning and pressurisation system, refers to the :

- ☐ means by which pressurisation is controlled.
- ☒ cold air unit (air cycle machine) arrangement.
- ☐ charge air across the inter-cooler heat exchanger.
- ☐ source of the charge air.

14

"Frequency wild" in relation to a AC generation system means the generator :

- ☒ output frequency varies with engine speed.
- ☐ voltage regulator is out of adjustment.
- ☐ output frequency is too low.
- ☒ output frequency is too high.

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15

If an altimeter indicates 3500 feet with QNH 1004.7 hPa set, what is the pressure altitude?

☐ 3500 ft

☐ 3160 ft

☒ 3755 ft

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16

An aircraft is flying at FL 290 at an RAS of 310 kt, temperature deviation -6°C, the LSS is:

☒ 583 kt

☐ 563 kt

☐ 596 kt

☐ 570 kt

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17

If an altimeter indicates 3500 FT with the actual QNH of 1004, 7 hPa set, the approximate pressure altitude is:

☐ 3500 FT

☐ 3255 FT

☒ 3745 FT

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18

During which stroke of a reciprocating engine is the gaseous mixture expanding within the cylinder?

☐ Compression.

☐ Intake.

☒ Power.

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19

For an aircraft flying at Mach 0.82 at FL 350, COAT - 35° C, the RAS would be:

☐ 251 kts

☒ 279 kts

☐ 264 kts

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20

An aircraft flying from warm air to cold air at a constant TAS. The Rectified Air Speed (RAS) would:

☒ **Increase**

☐ **Decrease**

☐ **Remain constant**

21

Considering an airspeed indicator, a second striped needle, if installed, indicates:

☐ VNE or VMO, depending on which is the higher.

☐ VMO.

☒ VNE or VMO, depending on which is the lowest.

☐ VNE.

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22

The elevation of aerodrome A is 390 feet and aerodrome B 450 feet. The pilot of an aeroplane at A, sets the altimeter to read aerodrome elevation and then flies to B without resetting the altimeter. Aerodrome pressure at A on departure was 1004 hPa. On landing at B the altimeter reads 630 feet, and assuming 1 hPa is 30 feet, the QNH is approximately:

☒ 1013 hPa

☐ 1019 hPa.

☐ 1011 hPa

23

When an altimeter is changed from 30.11" Hg to 29.96" Hg, in which direction will the indicated altitude change and by what value?

- ☐ Altimeter will indicate 15 feet lower.
- ☒ Altimeter will indicate 150 feet higher.
- ☐ Altimeter will indicate 150 feet lower.

24

Electrical bonding of an aircraft is used to: 1. protect the aircraft against lightning effects. 2. reset the electrostatic potential of the aircraft to a value approximating 0 volt 3. reduce radio interference on radiocommunication systems 4. set the aircraft to a single potential The combination regrouping all the correct statements

☒ 1, 2, 3

☐ 2, 4

☐ 1, 3, 4

☐ 3, 4

25

The position of the command bars of a flight director enables the pilot to know:

- 1 - the direction and the amplitude of the corrections to apply on the controls.
- 2 - only the direction of the corrections to apply on the controls.
- 3 - the attitude of the aircraft.

The combination that regroups all of the correct statements is:

☐ 2, 3.

☐ 1.

☒ 1, 3.

☐ 2.

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26

What is the movement of the center of pressure when the wingtips of a sweptwing airplane are shock-stalled first?

- ☒ Inward and forward.
- ☐ Inward and aft.
- ☒ Outward and forward.

27

A relay is :

- ☐ A switch specially designed for AC circuits.
- ☒ An electromagnetically operated switch.
- ☐ An electrical security switch.
- ☐ An electrical energy conversion unit.

28

A ring laser gyro can measure:

- ☒ *rotation about its sensitive axis.*
- ☐ *rotation in all directions.*
- ☐ *accelerations in all direction.*
- ☒ *accelerations about its sensitive axis.*

29

Running a fuel tank dry before switching tanks is not a good practice because

- ☒ **any foreign matter in the tank will be pumped into the fuel system.**
- ☐ **the engine-driven fuel pump or electric fuel boost pump draw air into the fuel system and cause vapor lock.**
- ☐ **the engine-driven fuel pump is lubricated by fuel and operating on a dry tank may cause pump failure.**

30

In a CADC the output of vertical speed is achieved by means of:

- ☒ a measurement of the rate of change of altitude as sensed by the static pressure transducer
- ☐ an integration of the CAS transducer output
- ☐ a dedicated vertical speed transducer
- ☒ a combination of the outputs of the dynamic and static pressure transducers

31

A passenger emergency mask is a :

- ☒ mask with flow on request and cannot be used if there is smoke in the cabin.
- ☐ continuous flow mask and can be used if there is smoke in the cabin.
- ☐ continuous flow mask and cannot be used if there is smoke in the cabin.
- ☐ mask with flow on request and can be used if there is smoke.

32

The vertical command bar of a flight director:

- 1 - repeats the position information given by the ILS in the horizontal plane
- 2 - repeats the position information given by the ILS in the vertical plane
- 3 - gives information about the direction and the amplitude of the corrections to be applied on the bank of the aircraft

The combination that regroups all of the correct statements is:

☐ 2, 3.

☐ 1.

☒ 3.

☐ 1, 3.

33

Which of these statements about structural design principles are correct or incorrect?

I. In structural design, fail safe implies parallel structural parts.

II. In structural design, safe life implies the structure will never fail during a declared time period or number of cycles.

☒ I is correct, II is incorrect.

☐ I is correct, II is correct.

☒ I is incorrect, II is correct.

☐ I is incorrect, II is incorrect.

34

The purpose of a diluter demand regulator in an oxygen system is to :

- ☒ is only recommended with smoke in the cockpit.
- ☐ deliver oxygen flow only above FL 100.
- ☐ deliver oxygen flow when inhaling.
- ☐ mix air and oxygen in a passenger oxygen mask.

35

The radio altimeter is required to indicate zero height AGL as the main wheels touch down on the runway. For this reason, it is necessary to:

- ☐ *adjust the gross height according to the aeroplane instantaneous pitch.*
- ☒ *have a specific radio altimeter dedicated to automatic landing.*
- ☐ *change the display scale in short final, in order to have a precise readout.*
- ☐ *compensate for residual height and cable length.*

36

For a turbine engine, the term self-sustaining speed means the engine

- ☐ is designed to idle after starting.
- ☐ operates most efficiently in the cruise..
- ☒ will run without any external assistance.
- ☒ will enable the generators to supply bus-bar voltage.

37

A gyroplane will have the greatest tendency to roll during

- ☒ horizontal flight at high speed.
- ☐ descending flight in which forward airspeed decreases.
- ☐ climbing flight in which forward airspeed decreases.

38

On an aeroplane, spoilers are :

- ☐ lower wing surface devices, their deflection is symmetrical or asymmetrical.
- ☒ upper wing surface devices, their deflection is symmetrical or asymmetrical.
- ☐ lower wing surface devices, their deflection is always asymmetrical.
- ☐ upper wing surface devices, their deflection is always asymmetrical.

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39

Pressure Altitude 8000 ft QNH Altitude 7500 ft OAT + 30° C Terrain elevation 5700 ft.
The approximate absolute altitude is:

☐ 500 ft

☒ 2650 ft

☐ 1800 ft

40

"Nose wheel shimmy" may be described as :

- ☒ the amount of free movement of the nose wheel before steering takes effect.
- ☐ a possibly damaging vibration of the nose wheel when moving on the ground.
- ☐ aircraft vibration caused by the nose wheel upon extension of the gear.
- ☐ the oscillatory movement of the nose wheel when extended prior to landing.

Attachment No. 1

Questions 6

Figure 84

instrument_84.jpg

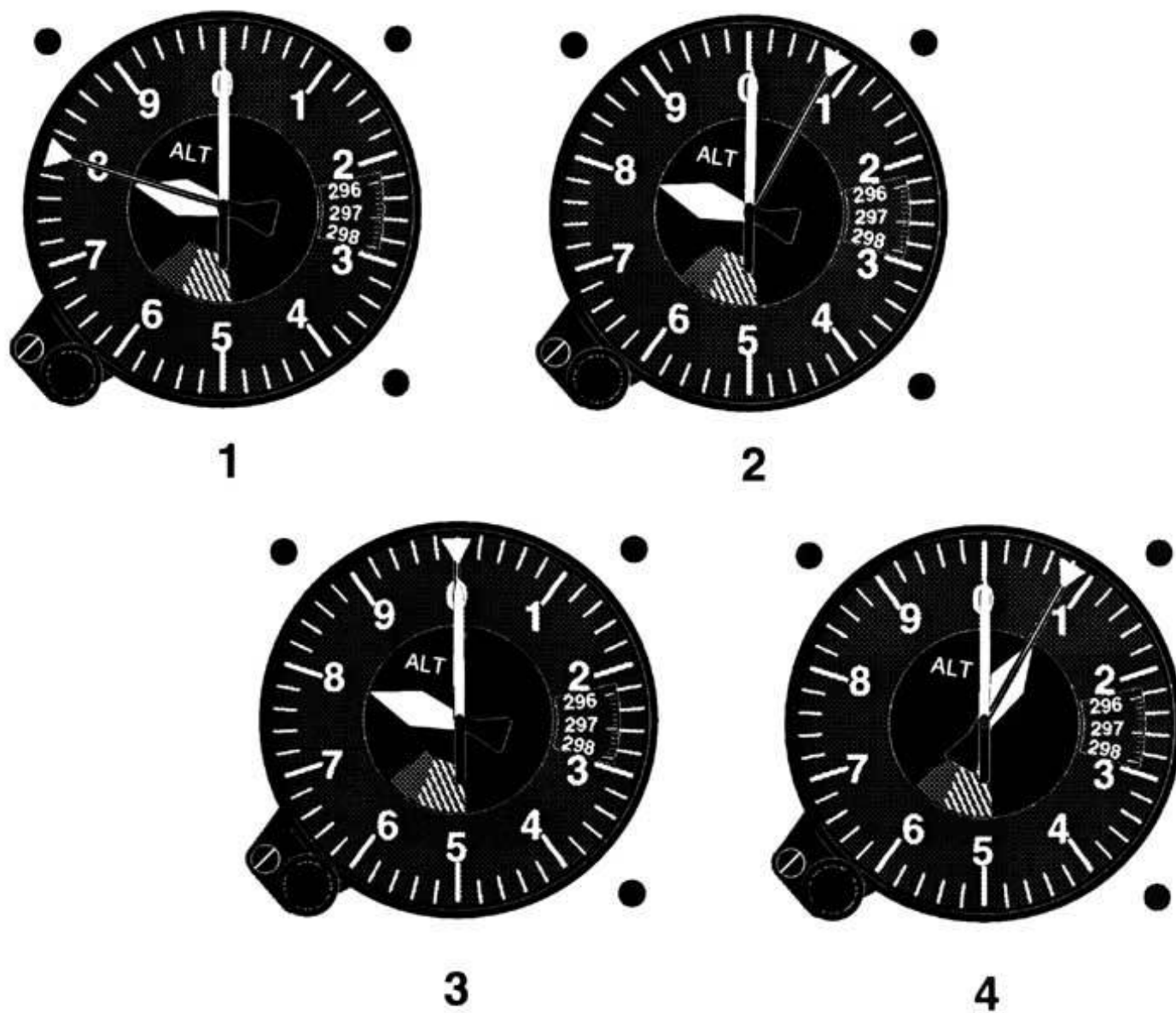


Figure 84. Altimeter/8,000 Feet © ASA